

Andrea Karlová

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[Website](#) | [GitHub](#) | [Google Scholar](#)

Professional Summary

- *PhD Candidate at UCL and Machine Learning Researcher with experience in applied mathematical modeling, deep learning, and computational science*
 - *Specialised in AI-driven drug discovery, generative modelling, and cheminformatics.*
 - *Experienced in building and guiding technical team and designing production ML pipelines on AWS.*
 - *Actively contributing to scientific progress through published research, securing InnovateUK grant, and continuing research work.*
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Academic & Research Experience

University College London (UCL)

Research Student, Computer Science | 2015 – Present

- **Drug Discovery Challenges:** Identified hits for [MurD Ligase](#) and the binding sites of SARS-CoV-2 ([nsp13](#) and [nsp3](#)). Utilized 3D interaction fingerprints and scoring methods in molecular dynamics (GROMACS), along with force-field calculations and recombination of molecular fragments.
- **Deep Learning & Generative Modelling:** Researched the robustness of latent representations using Optimal Transport Theory, VAEs, and GECO.
- **Collaborative Virtual Screening:** Collaborated with the University of Chemistry and Technology, Prague, to apply VAEs to molecular fingerprints (ECFP and FCFP), Reinforcement Learning for molecular design, docking (Vina), and E(n)-equivariant GNNs.
- **Materials Discovery:** Collaborated with the [University of Surrey](#) and UCL AI Centre on high-throughput perovskites photovoltaic discovery, evaluating atomic structures with GPAW, ASE, and RDKit.

Visiting Research Engagements (2010 – 2016)

- **UCL Computer Science (2015-2016):** Conducted research on IP-based financial instruments and trading platforms, applying Hidden Markov Models and RL market-making methods.
- **UCL Civil, Environmental & Geomatic Engineering (2015):** Evaluated the scalability of a high-volume Fusion and Analysis platform for geospatial clouds (co-funded by the EU Commission FP7 2002–2013, [IQmulus project](#)).
- **Imperial College London (2015-2016):** Worked jointly on term structure volatility surface models and fixed income models influenced by tempered Levy Flights.

- **Oxford-Man Institute & Columbia University (201)**: Created computational techniques and numerical approaches for option pricing models based on stable laws.
 - **Gorilla Science (2011–2016)**: Worked with [Dr. Patrick S. Hagan](#) on term structure volatility surface models using tempered Levy Flights to design robust hedges for forward-starting products.
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Industry & Applied R&D Experience

Matterhorn Studio

ML Research Scientist | Aug 2024 – Jun 2025

- Worked on Bayesian optimisation project in direct collaboration with **Novo Nordisk**.

London North Eastern Railway (LNER)

ML Technical Lead | Oct 2022 – Dec 2023

- **Leadership**: Built, hired, and led a senior technical team to deliver production-ready ML models; mentored data scientists and applied complete ML development lifecycle practices.
- **MLOps Infrastructure**: Created AWS architecture for the ML production pipeline and oversaw the setup of an ISO27007 compliant CI/CD ML platform.
- **Applied AI**: Directed projects such as fraud anomaly detection, customer propensity modeling, topic labeling, and train delay prediction.

April19 Discovery Ltd.

ML R&D Consultant | Jun 2021 – Sep 2022

- Specialised in molecular drug design targeting the central nervous system.
- Applied similarity searching, QSAR models, multi-task DNNs for target-assay pair prediction, bioactivity prediction, generative models, and 3D ligand-protein fingerprints.

Fractal Labs Ltd.

Data Science Lead / Senior Data Scientist | Jul 2018 – Aug 2020

- **Forecasting**: Created time-series forecasting models for SME cash flow using VAEs, time warping loss functions, Neural ODEs, CNN/RNN setups, Random Forests, and Kalman Filters.
- **Conversational AI**: Designed a domain-specific chatbot for SMEs using CNNs/RNNs, Q-learning with memory replay, and VAEs applied to language generation.
- **Leadership**: Set up a complete production pipeline with TensorFlow Extended and Docker. Served as Project Lead and successfully obtained a £300k InnovateUK grant.

StatusToday

Data Science and R&D | Nov 2017 – Apr 2018

- Created employee productivity analytics using randomised intensities of Compound Poisson processes and change points detection.

Quantitative Finance Roles (2005 – 2011)

- **PricewaterhouseCoopers (PwC) Czech Republic (2009 - 2011):** Consultant pricing exotic financial derivatives and illiquid products.
 - **KBC Global Services & CSOB (2005 - 2009):** Market Risk and Model Validation Associate managing methodologies and independent price verification for commodity trading and life insurance derivatives.
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Education

- **Pursuing PhD in Computer Science:** University College London, United Kingdom (2023 – Present)
 - **Pursued PhD in Mathematics:** Charles University, Prague, Czech Republic (Completed 2014; title unawarded due to supervisor's death)
 - **Mgr. (BSc + MSc equivalent) in Mathematics:** Charles University, Prague, Czech Republic (2005)
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Publications, Talks & Awards

Selected Publication:

1. (2026) Madhushika Silva et al.: *CACHE Challenge #4: Targeting the TKB Domain of the E3 ligase CBLB, an Immuno-Oncology Target*, *in preparation*, June 2026
2. (2026) Karlová A., Kabra R., Augusto de Souza D., Paige B.: [COBALT: Censored Optimization and Bayesian Active Learning Techniques](#), August 2026, UAI 2026
3. (2026) Herasymenko O. et al.: [CACHE Challenge #3: Targeting the Nsp3 Macrodomein of SARS-CoV-2](#), Jan 2026, J. Chem. Inf. Model., vol 66, issue 3, pp 1566–1581
4. (2025) Herasymenko O. et al.: [CACHE Challenge #2: Targeting the RNA Site of the SARS-CoV2 Helicase Nsp13](#), July 2025, J. Chem. Inf. Model., vol 65, issue 13, pp 6884 - 6898
5. (2024) Karlová A., Kabra R., Augusto de Souza D., Paige B.: [Variational Inference with Censored Gaussian Process Regressors](#), July 2024, ICML 2024 Workshop on Structured Probabilistic Inference & Generative Modeling
6. (2024) Karlová A., Paige B.: [SWUS: Active Learning with Structure Weighted Uncertainty Score](#), July 2024, ICML 2024 Workshop Efficient and Accessible Foundation Models for Biological Discovery
7. (2022) Karlová A., Dehaen W., Penciu A., Dallaway R., Goonatilake S.: [PEPSI: Post-docking Evaluation with Protein-Small Molecules Interaction](#), July 2022, Workshop on Computational Biology ICML 2022
8. (2021) Karlová A., Dehaen W., Penciu A.: [How to Reward Your Drug Agent](#), December 2021, Workshop on Deep Generative Models and Downstream Applications NeurIPS 2021
9. (2021) Karlová A., Underwood C., Silva R.: [Virtual Screening For Perovskites Discovery](#), July 2021, Workshop on Tackling Climate Change with ML, ICML 2021
10. (2021) Karlová A., Dehaen W., Svozil D.: [Molecular Fingerprint VAE](#), July 2021, Workshop on Computational Biology ICML 2021
11. (2020) Karlová A.: [Robustness of Latent Representations](#), July 2020, Conference Workshop Proceedings UDL ICML 2020
12. (2018) Klebanov L., Karlová A.: [Distributional Tail Estimation through its Characteristic Function](#), Mar 2018, Journal of Mathematical Sciences, Vol. 229, Issue 6, pp 714–718

13. (2016) Karlová A., Hagan P.S.; [On Beyond Black](#); Sept. 2016, Wilmott, p.42-69
14. (2013) Karlová A.; [On distribution of product of stable laws](#); SIAM Theory of Probability and its Applications (submitted September 2013, accepted July 2015)
15. (2008) A. Karlová; *Levy Processes and Balayage Spaces*; J. Antoch., G. Dohnal, (Eds.) ROBUST 2008, Proceedings of the 13th Summer School of the Union of Czech Mathematicians and Physicists, 171—176.

Selected Invited Talks

- AI UK (2022)
- QuantInsights (2019) & Women in Data Science (2019)
- 49th London PyData Meetup (2018)
- RISK Event Quant Summit Europe (2016)
- QMF Quantitative Methods in Finance, Sydney (2010)

Grants, Honours & Awards

- **Best Poster Award:** [Workshop on Computational Biology ICML \(2021\)](#).
- **InnovateUK Grant Awardee:** £300k secured as Project Lead (2019).
- **Marie Curie EU Grant Scholarships:** Funded participation in specialised stochastic modelling summer schools across Europe (2007, [2008](#)).

Teaching, Supervision & Academic Service

- **MSc Thesis Supervisor:** Co-supervised more than 10 MSc theses at UCL and the University of Bristol on subjects including Imitation Learning, Density Functional Theory, generative modelling, and virtual screening.
- **Teaching Assistant:** Worked as a TA for [Bayesian Deep Learning Methods](#) and gave [invited lectures](#) on Big Data processing, Docker, and Machine Learning pipelines at UCL.
- **Peer Reviewer:** Acted as a reviewer for *Nature Scientific Reports*, *NeurIPS*, *ICML*, *ICLR*, and *AAAI*.

Core Technical Skills

- **Programming Languages:** Python, C/C++, Bash, SQL.
- **ML & Deep Learning:** Geometric Deep Learning (E(n)-equivariant GNNs), Variational Inference (VAEs), Bayesian Optimisation, Reinforcement Learning, Neural ODEs, Optimal Transport Theory.
- **Computational Science:** Cheminformatics, Molecular Dynamics (GROMACS), Quantum Mechanical Computations (GPAW, ASE, RDKit).
- **Engineering & MLOps:** AWS Architecture, CI/CD (ISO27007 compliant), Docker, TensorFlow, PyTorch, MongoDB, Git, Linux (Debian)